



Reclaiming the Bus Stop

Illegal parking in bus laybys: the cost to On-Time Performance,
and a self-funding enforcement programme to fix it

Two problems, one parked car

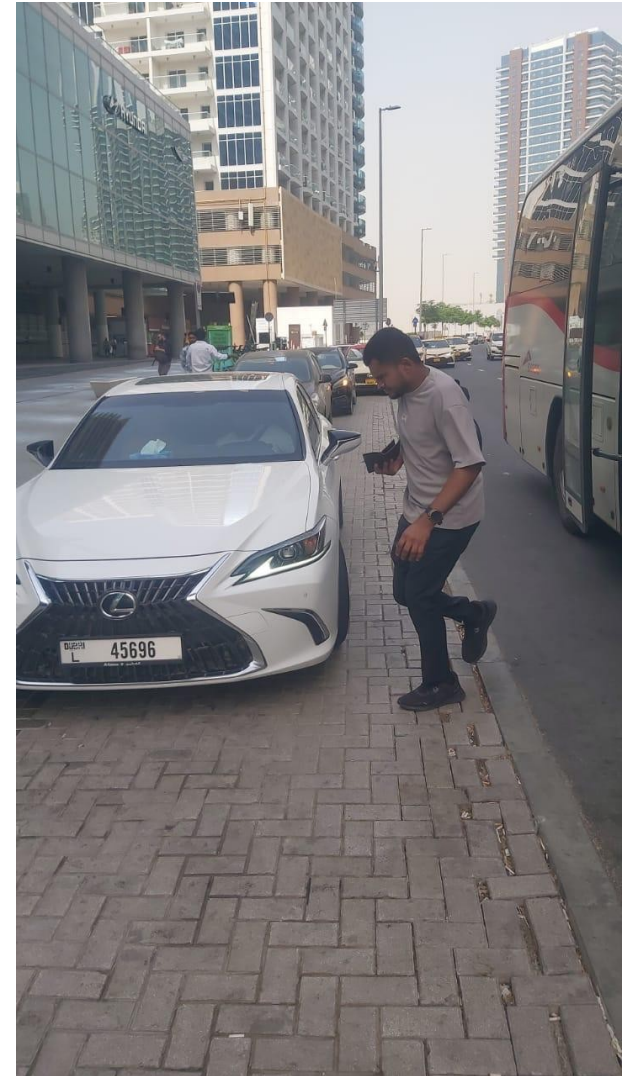
1 Buses can't reach the kerb

Taxis, limousines and private cars occupy bus laybys at busy stops. The bus queues, re-approaches or stops in the running lane — every event costs 30–60+ seconds and erodes On-Time Performance in the peak, exactly when the network is tightest.

2 Passengers board in live traffic

When the layby is blocked, doors open in the carriageway. Every blocked arrival is a safety and accessibility exposure — for the elderly, wheelchair users and children most of all.

This study quantifies both — from a field survey of 11 hotspots, one full day of AVL runtime data (75,780 stop records), and the network GTFS — and prices the fix.



Field photo, 28 Jul 2026: layby blocked, passengers board in the traffic lane

The evidence: one morning, 11 stops

60

blocked-layby cases in 3 hours
(07:00–10:00, one day)

1 in 21

AM-peak bus arrivals met
an obstructed layby

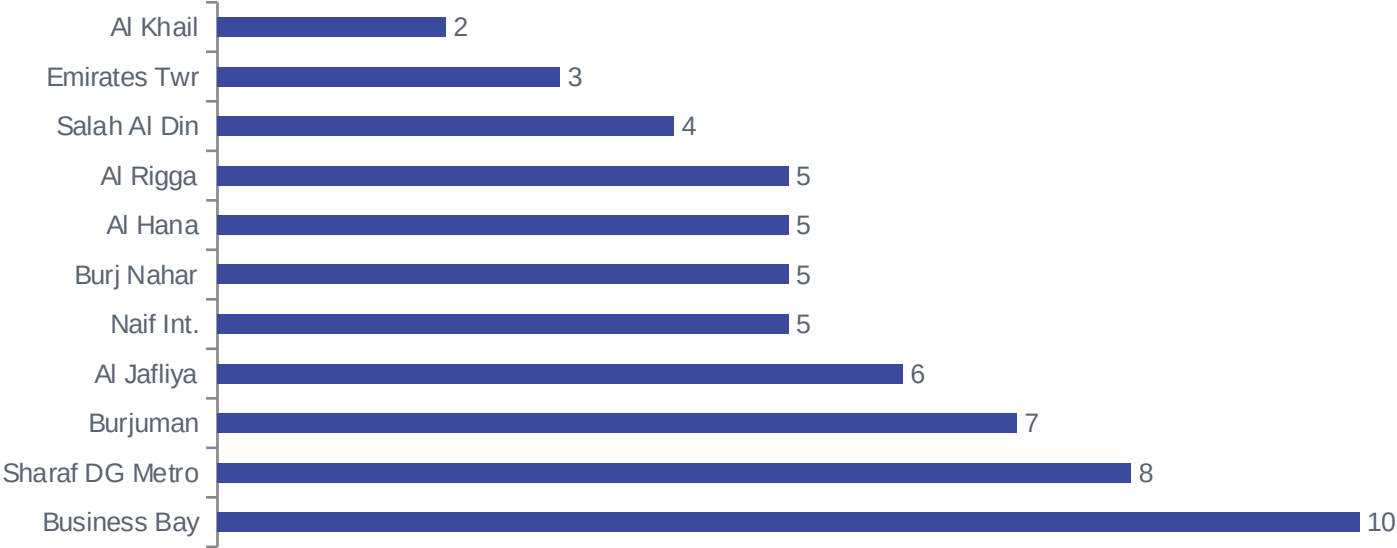
87.2%

baseline OTP that day
(38 routes, 72,679 stop events)

78.7%

OTP at Sharaf DG Metro —
worst stop, most blockages (8)

Blocked-layby cases by location (07:00–10:00, single day)



Cross-check

AVL data (22 Jul 2026) confirms these stops carry 1,254 bus arrivals in the AM peak alone — and that late running already concentrates in peak hours.

Network-wide: 912 stops carry the risk

Every stop in the GTFS network was scored on the surveyed hotspot profile: peak bus throughput plus metro-station or mall adjacency.

Tier	Stops	Definition	Share of peak arrivals	Annual blockages
High	112	Matches surveyed profile (metro/mall + heavy throughput) — survey-calibrated rate	24%	≈ 182,000
Medium	400	Next 400 by peak throughput — 50% of survey rate (assumption)	36%	≈ 134,000
Low	400	Following 400 — 25% of survey rate (assumption)	16%	≈ 30,000
Total	912	4 in 10 network stops	76%	≈ 346,000

76%

of all peak bus arrivals happen at these 912 stops

7.5

violations per stop per day at High-tier stops

Biggest non-surveyed exposures: Ibn Battuta, Union Metro, Mall of the Emirates, Gold Souq, Dubai Mall Metro.

What it costs the network every year

346,000

bus arrivals obstructed per year
(912 stops, both peaks)

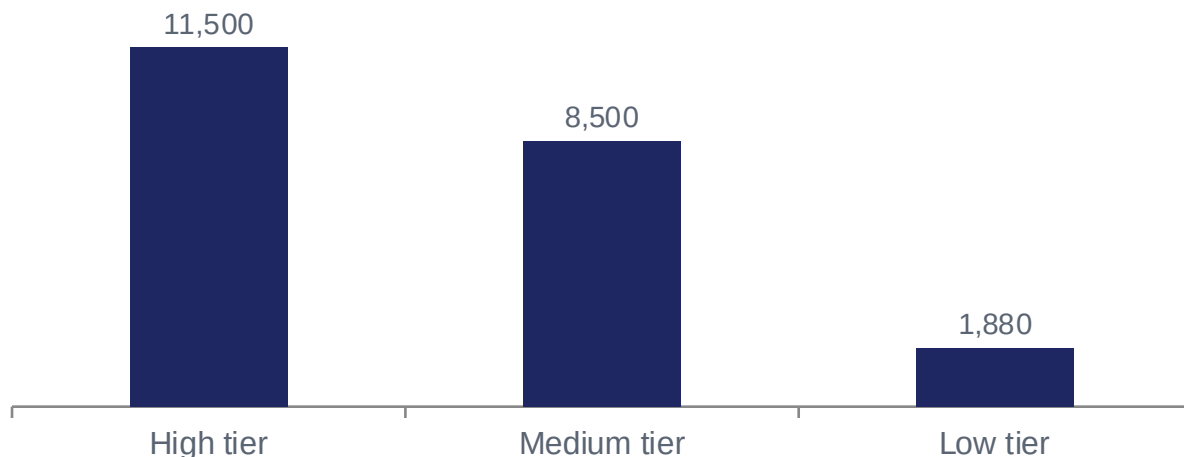
2,900 h

of direct bus delay per year
(at 30 s per obstruction)

−0.85 pp

peak OTP lost network-wide
(30–60 s scenarios: −0.35 to −0.85)

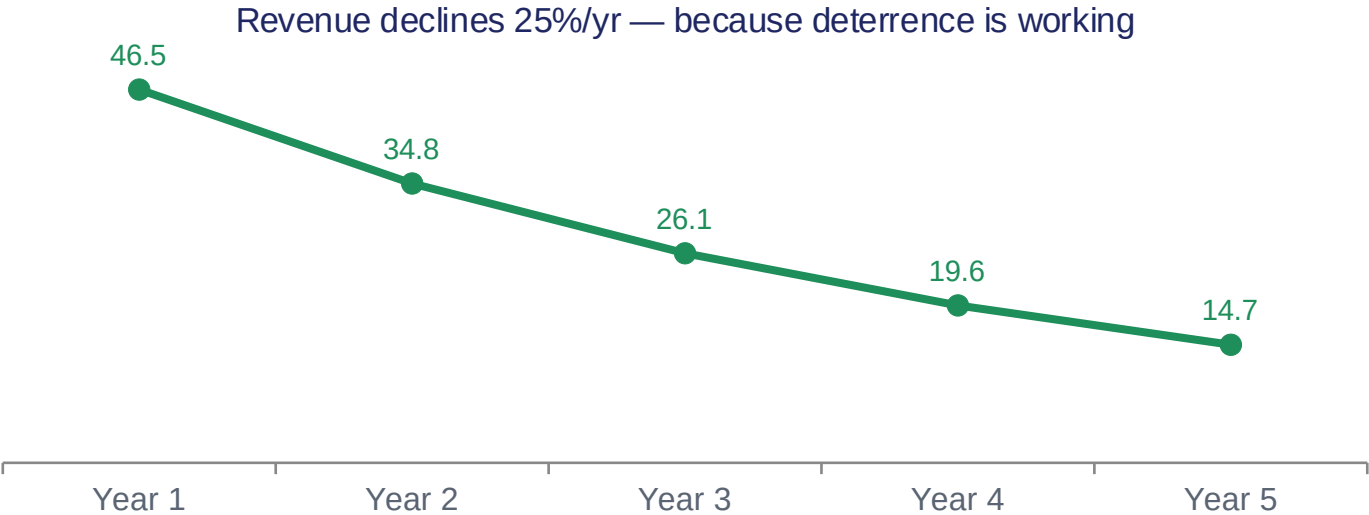
Extra late trips per year by tier (60 s per obstruction)



Why small delays matter

OTP counts a trip late beyond +5 minutes. The damage lands on trips already 4–5 minutes behind — 12.7% of peak stop events. A 30-second obstruction tips 1 in 38 of the trips it touches over the line; at 60 seconds, 1 in 16.

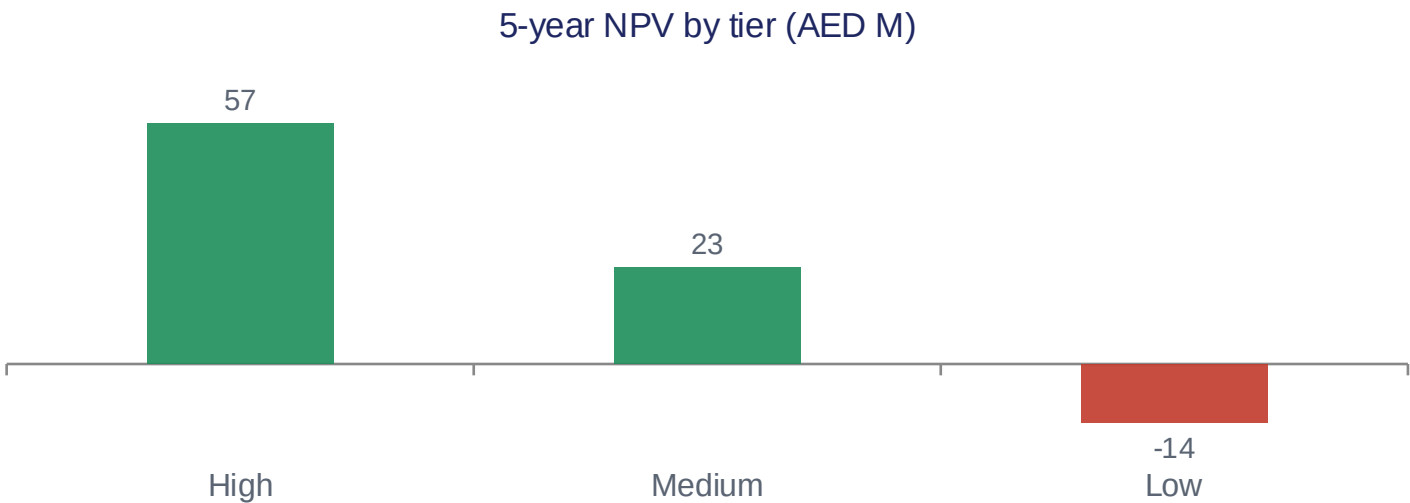
Enforcement pays: the violation-to-cash funnel



The honest pitch
Fines finance the system in years 1–3. The permanent return is OTP recovery and the end of live-lane boarding.

Option A — fixed cameras: invest where density pays

Tier	Capex	Revenue yr 1	5-yr NPV @7%	Payback
High — 112 stops	AED 4.5M	AED 24.4M	+56.6M	< 1 year
Medium — 400 stops	AED 16.0M	AED 18.0M	+22.6M	~1 year
Low — 400 stops	AED 16.0M	AED 4.0M	-13.8M	never



The verdict

Stop at High + Medium.

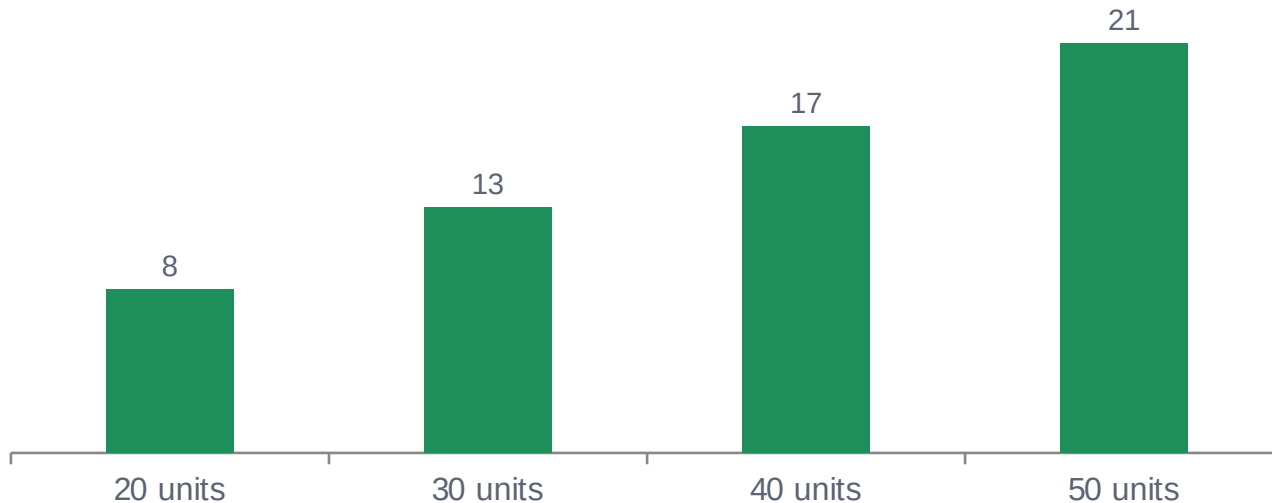
512 stops · AED 20.5M capex
5-yr NPV +79.2M · payback < 1 yr

The Low tier's violation density is too thin to carry a fixed installation — leave it to the portable fleet.

Assumed: AED 40k/site capex, 5k/yr opex (benchmark; pre-RFP).

Option B — portable cameras: small capex, follows the problem

5-yr NPV by fleet size, monthly rotation at High-tier stops (AED M)



7.5

violations seen per
deployed camera per day

3.5×

daily revenue vs daily
all-in cost per unit

Deployed at High-tier stops; deterrence decay 15%/yr under monthly rotation.

Economics per unit: $\approx$ AED 830/day collected vs $\approx$ AED 240/day cost (AED 15k capex + 60k/yr crew, vehicle, redeployment). Every fleet size is NPV-positive — value scales with units, and rotation adds a "cameras could be anywhere" deterrence halo that fixed sites lack. Trade-off: only 2–5% of stop-time covered, so OTP recovery is slower than fixed coverage.

Recommendation: the hybrid programme

Phase 1 · Months 0–6

Fixed cameras at 112 High-tier stops + 20 portable units.
Run the validation survey that firms Medium/Low rates.
Capex ≈ AED 4.8M.

Phase 2 · Months 6–18

Extend fixed coverage to the 400 Medium-tier stops.
Grow portable fleet to 30–50 as validated data directs.
Capex ≈ AED 16M — funded by Phase 1 revenue.

Phase 3 · Year 2+

Portable rotation covers the Low tier and emerging hotspots.
Integrate with the fines system; publish OTP recovery.
Review AED 200 vs AED 500 federal fine level.

+79M

5-yr NPV, fixed High+Medium (AED)

+0.85pp

peak OTP recoverable network-wide

< 1 yr

payback on every element deployed

Revenue is the financing bridge, not the goal: it declines as compliance improves, while the OTP and safety gains compound.

Every peer city has already moved

New York — MTA ACE

Bus-mounted cameras, escalating fines US\$50–250

~5% faster buses, ~20% fewer collisions, ~30% fewer repeat violations

London — TfL

Bus stop clearways, CCTV-enforced 24/7, PCN £160

Near-universal kerb access; compliance underpins the iBus OTP regime

Singapore — LTA

Camera-equipped buses since 2008, ~S\$150 fine

Sustained bus-lane and stop compliance

Sydney — TfNSW

Bus-zone offence ≈ A\$390

Deterrence through penalty level

Dubai today: manual patrols only. This study models a conservative AED 200 fine — the federal schedule already allows AED 500 for stopping in a bus stop, which would scale every revenue figure ×2.5 and turn even the Low tier positive.

Decisions requested today

1

Approve Phase 1

AED ~5M: fixed cameras at the 112 High-tier stops plus a 20-unit portable fleet. Self-funding within year 1.

2

Confirm revenue retention

Agree with Dubai Police / DoF how bus-stop fine revenue is shared — the single biggest swing factor in the cash ROI.

3

Commission the validation survey

1–2 weeks at ~20 Medium/Low-tier stops to replace the two assumed rate multipliers with measurements.

4

Issue the enforcement RFP

ANPR fixed sites + portable units + fines-system integration, structured to allow bus-mounted cameras (NYC model) as a Phase 3 extension.